

Phanendra Paladugu

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SUMMARY

Data scientist with an M.S. in Electrical and Computer Engineering and 4 years of applied machine learning, focused on building production LLM systems for engineering content. Built RAG, document processing, and agentic workflows (LangGraph, LangChain, ChromaDB, pgvector) with measured retrieval quality, and fine-tuned and benchmarked a 7B model that beats GPT-4o on BIRD text-to-SQL. Applies ML to semiconductor manufacturing data (SECOM wafer fail detection) and works daily with agentic coding tools such as Claude Code. Ships reliable, traceable services on Docker, Kubernetes, AWS, and GCP.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

GenAI & NLP: Hugging Face Transformers, LLM fine-tuning (QLoRA, GRPO), quantization (GGUF), RAG, LangChain, LangGraph, vector search (FAISS, ChromaDB, pgvector), SpaCy, LLM-as-judge evaluation, prompt engineering, agentic workflows, AI coding assistants (Claude Code)

Machine Learning: Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, anomaly detection, time series forecasting, classification, hyperparameter tuning, Explainable AI (SHAP, LIME)

MLOps & Deployment: FastAPI, Flask, Docker, Kubernetes, MLflow, Evidently AI, Prometheus, Grafana, GitHub Actions, Git, Streamlit

Cloud: AWS (S3, EC2, SageMaker), GCP (BigQuery, Vertex AI)

Analytics & Visualization: Pandas, Scikit-learn, Matplotlib, Power BI, Streamlit, SHAP

PROFESSIONAL EXPERIENCE

Data Scientist

Sep 2026 – Present

Axitem Software Solution Inc, Livonia, MI (Remote)

- Design, develop, and deploy machine learning, predictive analytics, and statistical models that support strategic business decisions.
- Collect, clean, and analyze structured and unstructured data with Python and SQL; perform exploratory data analysis, feature engineering, and statistical modeling to surface actionable trends.
- Build and maintain data pipelines and automated workflows for ingestion, model training, and deployment; validate models with performance metrics, tests, and monitoring.

Data Scientist (Contract)

Jan 2025 – Jul 2026

Wells Fargo, Remote

- Built NLP and RAG pipelines for compliance document classification, entity extraction, and search with SpaCy, Hugging Face Transformers, LangChain, and ChromaDB; measured retrieval quality with Recall@K and MRR.
- Built transaction anomaly detection and fraud models on banking data with Python, Scikit-learn, and TensorFlow, owning feature design and model evaluation through design review.
- Deployed models as JWT-secured FastAPI services on Docker and Kubernetes with logging, health checks, and unit tests; tracked experiments in MLflow and drift with Evidently AI.
- Wrote Python and SQL pipelines to extract data, run automated validation checks, and assemble training and test datasets for scheduled model runs.

Lab Assistant, Electrical Engineering

Aug 2024 – Dec 2024

Purdue University Northwest, Hammond, IN

- Instructed students in electrical measurements (voltage, current, impedance, frequency response) using oscilloscopes, multimeters, and signal generators across multiple lab modules.
- Reviewed lab notebooks and technical reports for measurement accuracy, documentation quality, and analytical reasoning.

Technology Analyst

Jan 2020 – Dec 2021

Infosys, Pune, India

- Designed and deployed a TensorFlow/Keras deep learning classifier with NLP text similarity to auto-categorize 200+ monthly incidents, cutting resolution time by 30%.
- Developed anomaly detection and demand-forecasting models for supply chain data using ensemble methods and time series analysis.
- Built end-to-end ML pipelines (preprocessing, feature engineering, training, hyperparameter tuning with GridSearch/RandomSearch, evaluation) using Scikit-learn and XGBoost.

- Deployed models as Flask and FastAPI REST APIs with logging and monitoring, and built Power BI dashboards to surface model performance for stakeholders.

Senior Systems Engineer

Jan 2017 – Dec 2019

Infosys, Pune, India

- Automated data integration and ETL workflows using Python, Selenium, and Sikuli across SAP BW/APO, saving 800+ hours annually.
- Managed data integrity across SAP APO and MM, running data-quality audits and Python validation scripts to detect anomalies.
- Built structured data pipelines and wrote complex SQL for extraction, transformation, and validation supporting global operations.

PROJECTS

OmniAsk: Agentic RAG and Text-to-SQL Assistant

github.com/phanipal/omniask

- Built a LangGraph agent routing questions to read-only text-to-SQL, document RAG, or human-approved actions; served over streaming FastAPI on Kubernetes with Prometheus/Grafana/Langfuse observability.

DocBrain: Document Intelligence Engine with Quantitative Evaluation

github.com/phanipal/docbrain

- Built a grounded retrieval pipeline over 4 document types with quality-based OCR routing (Tesseract/DocTR), SpaCy NER, and ChromaDB retrieval, backed by 162 pytest tests.
- Evaluated on 30 hand-written queries (0.90 Recall@5, 0.78 MRR, 1.00 Recall@1 on contracts) plus an LLM-as-judge faithfulness score.

Slonik-7B-GRPO: Text-to-SQL Fine-Tuned LLM

huggingface.co/Phani-labs/Slonik-7B-GRPO

- Fine-tuned Qwen2.5-Coder-7B with QLoRA and execution-based reinforcement learning (GRPO); raised BIRD (PostgreSQL split) accuracy from 12.22% to 38.20%, outperforming GPT-4o (34.44%), and published GGUF quantizations for local deployment.

SECOM Wafer Fail Detection: Sensor-Based Anomaly Detection

github.com/phanipal/Secom_Anomaly_Detection

- Screened 1,567 wafer lots with 590 process sensor readings (6.6% fail rate); filtered to 446 usable sensors and compared Isolation Forest, class-weighted Logistic Regression, and XGBoost under leakage-safe stratified 5-fold cross-validation.
- Class-weighted XGBoost reached 0.71 ROC-AUC and 0.17 PR-AUC (2.5x the random baseline), catching 34% of failing lots at a 10% false-alarm rate; ranked the 10 sensors driving fail risk with SHAP and scheduled the pipeline as a daily Apache Airflow DAG with a PR-AUC floor that blocks bad retrains.

EDUCATION

Master of Science, Electrical and Computer Engineering | GPA 3.61/4.0

Aug 2023 – May 2025

Purdue University Northwest, Hammond, IN

Coursework: Digital Signal Processing, Machine Learning, Neural Networks, Computer Vision, Big Data, Computer Networks

Bachelor of Technology, Electrical, Electronics and Communications Engineering

2012 – 2016

Vasireddy Venkatadri Institute of Technology, Guntur, India